

Time Series

Applied Time Series Analysis STA457 Lecture Notes

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Time Series Analysis

More on AR(1) Processes

Lemma (Properties of AR(1) Model)

Let us assume $\{W_t\}$ is a purely random process, e.g., $W_1, W_2, \dots \sim \text{wn}(0, \sigma_w^2)$. Then, for the first order autoregressive process defined by

$$AR(1) : X_t = \delta + \phi X_{t-1} + W_t, \quad (1)$$

the following properties

- ❶ $\mathbb{E}[X_t] = \frac{\delta}{1 - \phi}$
- ❷ $\text{Var}[X_t] = \frac{\sigma_w^2}{1 - \phi^2}$
- ❸ $\gamma_{t,t+h} = \frac{\sigma_w^2 \phi^h}{1 - \phi^2}$
- ❹ $\rho_{t,t+h} = \phi^h$

hold true if $|\phi| < 1$.

Properties of AR(1) - First Proof

Let us assume that for $t = t_0$ the initial value X_{t_0} is given. By successive substitution in equation of AR(1), we get

$$X_{t_0+1} = \delta + \phi X_{t_0} + W_{t_0+1}$$

$$\begin{aligned} X_{t_0+2} &= \delta + \phi X_{t_0+1} + W_{t_0+2} \\ &= \delta + \phi(\delta + \phi X_{t_0} + W_{t_0+1}) + W_{t_0+2} \\ &= \delta + \phi\delta + \phi^2 X_{t_0} + \phi W_{t_0+1} + W_{t_0+2} \end{aligned}$$

$$\begin{aligned} X_{t_0+3} &= \delta + \phi X_{t_0+2} + W_{t_0+3} \\ &= \delta + \phi(\delta + \phi X_{t_0+1} + W_{t_0+2}) + W_{t_0+3} \\ &= \delta + \phi\delta + \phi^2\delta + \phi^3 X_{t_0} + \phi^2 W_{t_0+1} + \phi W_{t_0+2} + W_{t_0+3} \end{aligned}$$

$$\vdots$$

$$\begin{aligned} X_{t_0+h} &= (1 + \phi + \phi^2 + \cdots + \phi^{h-1})\delta + \phi^h X_{t_0} + \\ &\quad (\phi^{h-1} W_{t_0+1} + \phi^{h-2} W_{t_0+2} + \cdots + \phi W_{t_0+h-1} + W_{t_0+h}) \end{aligned}$$

OR

$$X_{t_0+h} = \phi^h X_{t_0} + \frac{1 - \phi^h}{1 - \phi} \delta + \sum_{j=0}^{h-1} \phi^j W_{t_0+h-j}. \quad (3)$$

Properties of AR(1) - First Proof (Cont'd)

For $t = t_0 + h$, we then get

$$X_t = \phi^{t-t_0} X_{t_0} + \frac{1 - \phi^{t-t_0}}{1 - \phi} \delta + \sum_{j=0}^{t-t_0-1} \phi^j W_{t-j}. \quad (4)$$

We can also assume that a fixed initial condition is given as X_0 at time $t_0 = 0$. Then, the formula simplifies as follows:

$$X_t = \phi^t X_0 + \frac{1 - \phi^t}{1 - \phi} \delta + \sum_{j=0}^{t-1} \phi^j W_{t-j}. \quad (5)$$

Since $|\phi| < 1$, for $t \rightarrow \infty$, we get

$$X_t = \frac{\delta}{1 - \phi} + \sum_{j=0}^{\infty} \phi^j W_{t-j}. \quad (6)$$

So,

$$\mathbb{E}[X_t] = \frac{\delta}{1 - \phi} + \sum_{j=0}^{\infty} \phi^j \mathbb{E}[W_{t-j}], \quad (7)$$

which yields

$$\mathbb{E}[X_t] = \frac{\delta}{1 - \phi}. \quad (8)$$

Properties of AR(1) - First Proof (Cont'd)

The autocovariance function is further obtained as follows:

$$\gamma_{t,t+h} = \text{Cov}(X_t, X_{t+h}) = \sigma_w^2 \sum_{j=0}^{\infty} \phi^j \phi^{j+h} = \sigma_w^2 \phi^h \sum_{j=0}^{\infty} \phi^{2j} = \frac{\sigma_w^2 \phi^h}{1 - \phi^2}. \quad (9)$$

Then, the variance of AR(1) is

$$\gamma_{t,t} = \text{Var}[X_t] = \frac{\sigma_w^2 \phi^0}{1 - \phi^2} = \frac{\sigma_w^2}{1 - \phi^2}. \quad (10)$$

Hence, the autocorrelation coefficient function of AR(1) is

$$\rho_{t,t+h} = \frac{\gamma_{t,t+h}}{\gamma_{t,t}} = \phi^h. \quad (11)$$

It is seen that the mean value of an AR(1) process is a constant. Furthermore, the autocovariance and autocorrelation coefficient functions depend only on the lag parameter h (not on both time t and h).

Lag or Backshift Operator

- 1 The *lag (or backshift) operator*, denoted by L (or B), is defined as follows:

$$L X_t = X_{t-1}. \quad (12)$$

- 2 The lag operator can be raised to powers, e.g.,

$$L^2 X_t = L(L X_t) = L X_{t-1} = X_{t-2}, \quad (13)$$

and in general

$$L^p X_t = X_{t-p}. \quad (14)$$

- 3 We can form *polynomials* using the lag operator, i.e.,

$$a(L) = a_0 + a_1 L + a_2 L^2 + \cdots + a_p L^p, \quad (15)$$

whose application on $\{X_t\}$ results in

$$a(L)X_t = a_0 X_t + a_1 X_{t-1} + a_2 X_{t-2} + \cdots + a_p X_{t-p}. \quad (16)$$

- 4 *Lag polynomials* can be multiplied, where multiplication is commutative, i.e.,

$$a(L)b(L) = b(L)a(L). \quad (17)$$

- 5 Some lag polynomials can be *inverted*. If $|\phi| < 1$, we define $(1 - \phi L)^{-1}$ as follows:

$$\frac{1}{1 - \phi L} = \sum_{j=0}^{\infty} \phi^j L^j. \quad (18)$$

Properties of AR(1) - Second Proof

Using the lag operator L , the AR(1) process can be rewritten as follows:

$$AR(1) : (1 - \phi L)X_t = \delta + W_t. \quad (19)$$

Solving for X_t , we then get

$$X_t = \frac{\delta}{1 - \phi L} + \frac{1}{1 - \phi L} W_t, \quad (20)$$

which can formally be expanded to a geometric series, yielding

$$\begin{aligned} X_t &= (1 + \phi L + \phi^2 L^2 + \dots) \delta + (1 + \phi L + \phi^2 L^2 + \dots) W_t \\ &= (1 + \phi + \phi^2 + \dots) \delta + (W_t + \phi W_{t-1} + \phi^2 W_{t-2} + \dots). \end{aligned}$$

Since $|\phi| < 1$, we then obtain

$$X_t = \frac{\delta}{1 - \phi} + \sum_{j=0}^{\infty} \phi^j W_{t-j}. \quad (22)$$

Due to representation above, we get for the mean

$$\mathbb{E}[X_t] = \frac{\delta}{1 - \phi} + \sum_{j=0}^{\infty} \phi^j \mathbb{E}[W_{t-j}] = \frac{\delta}{1 - \phi}. \quad (23)$$

Properties of AR(1) - Second Proof (Cont'd)

Properties of AR(1) - Second Proof (Cont'd)